

Chapter 5 - Practice Set

Lists

Write each program below as its own separate Python file (for example, q1.py, q2.py, and so on) and run it to check your work.

Question 1

Write a program that stores a list of your five favorite movies and prints: the full list, the first movie in the list, the last movie in the list, and the total number of movies using `len()`.

Question 2

Write a program that stores the list `[10, 20, 30, 40, 50]` in a variable and prints it. Then change the value at index 2 to 99 and print the list again, to demonstrate that lists can be modified after creation.

Question 3

Write a program that stores a list of eight numbers of your choice and prints: the first three numbers, the last three numbers, the whole list reversed, and every second number, all using slicing.

Question 4

Write a program that stores a list of grocery items, then uses `append()` to add one new item to the end, `insert()` to add another item at index 1, and `pop()` to remove the last item from the list. Print the list after each of these three operations.

Question 5

Write a program that stores the list `[5, 10, 15, 10, 20]` in a variable. Use `remove()` to delete the first occurrence of the value 10, then print the list. Separately, use `pop()` with an index to delete the element at position 0, and print the list again.

Question 6

Write a program that stores an unsorted list of at least six numbers. Print the list sorted in ascending order using `sort()`, then print it sorted in descending order using `sort(reverse=True)`, and finally print it reversed using `reverse()`.

Question 7

Write a program that stores a list of usernames allowed to log in to a system. Ask the user to type a username using `input()`, then use the `in` and `not in` operators to print a message stating whether or not that username is on the allowed list.

Question 8

Write a program that creates a nested list representing a 3x3 grid of numbers (three inner lists, each containing three numbers). Print the entire grid, print one full row using a single index, print one individual number using double indexing, and then modify one number in the grid and print the updated grid.

Question 9

Write a program that stores a list of at least five student names and uses a `for` loop with `enumerate()` to print each student's position (starting from 1) alongside their name, in the format "1. <name>".

Question 10

Write a program that stores a list of numbers from 1 to 20 and uses a single list comprehension to create a new list containing only the square of each EVEN number in the original list. Print the resulting list.